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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Team Name: *Idealogists*

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Problem Statement

Fishermen from Tamil Nadu (especially from coastal areas like Rameshwaram, Nagapattinam, etc. . .) sometimes cross the India – Sri Lanka Internation Maritime Boundary Line (IMBL). This may be due to:

- Depleting fish stock in Indian waters, due to which they enter Sri Lankan waters
- Wind and currents - accidental crossing
- Not being aware of the IMBL since many boats do not have GPS

Once they cross the border, intentionally or unintentionally, the Sri Lankan Navy arrests them and seizes their boats. This causes suffering and distress to the fishermen and their families.

Problem Understanding

- Who experiences this problem?
  - Fishermen from areas near the Palk Bay, such as Rameshwaram, Pamban and Nagapattinam
  - The family of fishermen, since an arrest of a fisherman can remove the household's only income source.

- What causes the problem?
  - Declining Fish Stocks in Indian Waters: Decades of fishing have depleted juvenile fish and damaged seabeds.
  - Difficulty Identifying the IMBL: Invisible at sea, and most small boats lack reliable boundary alerts. Offshore mobile networks are unreliable. Night fishing increases the risk of accidental crossing.

- How does it affect users or stakeholders?
  - Arrests are frequent. As of 9 October 2025, 74 fishermen and 242 boats from Tamil Nadu were still in Sri Lankan custody, according to the Chief Minister's letter to the External Affairs Minister.
  - The result is lost income, debt from lost boats and long separations from families.

Existing Gaps & Challenges

- Limitation 1: Border alerts are available on very few boats
- Limitation 2: The systems are complex and don't warn step by step
- Main Challenge: Arrests continue despite these devices, because an alert doesn't fix the reason for crossing. A solution needs to be cheap enough for every boat, work offline with no network, warn early in Tamil, and be paired with fishing-ground guidance that gives fishermen a reason to stay on the Indian side.

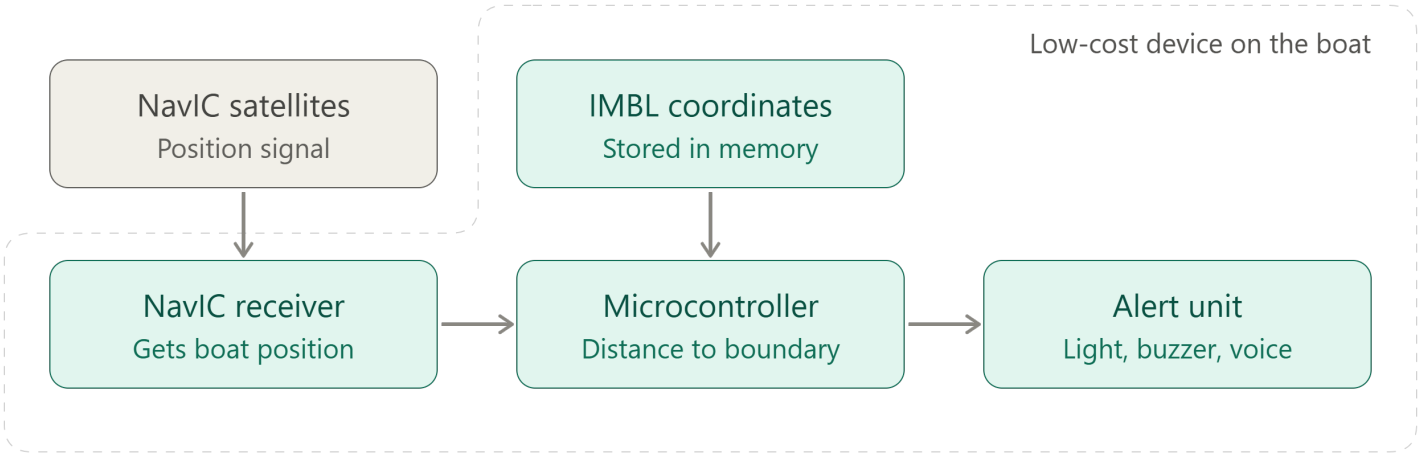
Our Proposed Solution

- Core Idea: A low-cost, NavIC-based boundary alert device for fishing boats. It tells the fisherman, step by step, when the boat is getting close to the IMBL, before the line is crossed.

- How the solution works at a high level:
  - The device uses a NavIC receiver to get the boat's position.
  - The IMBL coordinates are stored inside the device, so it compares the boat's position with the boundary on its own, without needing mobile network or internet.
  - As the boat gets closer, the alert gets stronger, from a green light to a warning buzzer to a loud alarm with a voice message in Tamil.

- Value delivered to users: Fishermen get an early warning (in Tamil), even far from the coast, so they can avoid accidental crossing and protect their boats and income. It is low-cost enough so that small boats can also afford it.

Technical Approach



The NavIC receiver gives the boat's position, the microcontroller checks its distance from the stored IMBL, and the alert unit warns the fisherman in stages.

Tech Stack

Frontend: LED, buzzer, voice alert

Backend: C/C++ firmware (ESP32)

Database: IMBL coordinates database

AI/ML, APIs, or Hardware: NavIC module, voice module

Innovation & Uniqueness

- What is new or different? Most existing devices are made for emergencies and weather alerts. Ours is built only for the boundary and warns step by step as the boat gets closer, with a light, then a buzzer, then a Tamil voice message, instead of one message.

- Why is this approach suitable? Fishermen go far offshore where there is no mobile network or internet, so the device keeps the boundary stored inside it. It also needs no phone, app or control station, which makes it simple to use and cheap enough for small boats.

- What is the key differentiator? It is a low-cost, offline, Tamil-language early warning made specifically for the India–Sri Lanka boundary, so every small boat can have one, not just boats that can afford satellite-solutions

Feasibility & Scalability

- **Feasibility:** The device a NavIC receiver module, a microcontroller, a buzzer and a voice module, so it can be built as a prototype with basic electronics skills. The steps are to build the prototype, store the IMBL coordinates, test the alert distances in a simulation, then run a field trial with a small group of fishermen.

- **Scalability:** The same device works for any boat, so it can be scaled to the whole Tamil Nadu coast. The stored boundary can also be updated for other sea borders, such as Gujarat's coast.

- **Risks:** GPS signal errors near the boundary, saltwater and battery damage, fishermen not trusting or using the device. We reduce them with a safety buffer before the line, waterproof casing with a long-life battery, and training in Tamil.

Cost for Product

- Development or hardware cost: About Rs.2,500 per device (NavIC receiver: Rs.1000, ESP32: Rs.450, Battery: Rs.400, Waterproof Casing: Rs.300, Speaker + Buzzer + LEDs: Rs.300)
- Hosting or maintenance cost: No hosting cost, very minimal maintenance cost
- Estimated total: Rs.2500 per device, where Rs. is Indian Rupee (INR)

Use Cases:

- Case 1: Fisherman, night fishing near Rameshwaram: The boat drifts with the current toward the boundary in the dark. The device shows a light at the caution distance, sounds the buzzer when closer, and plays a Tamil voice warning near the line, so the fisherman turns back before crossing.

- Case 2: Coast Guard and the fisheries department: With devices on many boats, fewer accidental crossings happen, so fewer rescue and release operations are needed. The same device design can also be updated with boundary coordinates for other coasts, such as Gujarat.

- Case 3: Fisherman with a poor catch, chasing fish toward the boundary: Catch near the Indian coast is low, so a fisherman moves further out in search of fish, close to the boundary, and may not realise how near he is. As he moves closer to the boundary, the device warns him in stages: first a light, then a buzzer, then a Tamil voice message. He can then stop and fish elsewhere on the Indian side instead of crossing.

Impacts and Benefits

- Benefits to the target users: Fishermen get an early warning in Tamil, even far from the coast, so they can turn back before crossing the boundary. This protects their freedom, boats and income.

- Social, Economic and Environmental Impact: Fewer arrests means fewer families separated and less distress in fishing villages. Economically, fewer seized boats and fewer lost fishing days mean less debt, and less money spent by governments on rescue efforts. Environmentally, if the alert is paired with fishing-zone guidance, boats can be steered away from overfished boundary waters.

- Expected Measurable Outcome: We fit devices on 50 boats in one harbour and count how many of those boats get arrested in one fishing season. Our goal is zero, and we compare it with last season's arrests from the same harbour.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 1	Target 1.5	No Poverty	Protects poor fishing families from sudden income loss and debt caused by arrests and lost boats.
SDG 14	Target 14.B	Life Below Water	Helps fishermen keep their access to the sea and their fishing livelihood by avoiding arrests and boat seizures.

Individual Member Contribution

Member	Contribution or responsibility
Goppamagari Manu Deep Reddy	Team Leader, managed the whole team, took care of Yukthi Portal Upload Process completely
P Nishanth	Took care of presentation (both pre-ideathon and post-ideathon)
P Dhivyes	Took care of chartwork, ideation process writing
V Ragghul	Took care of chartwork, ideation, ensured our thoughts converted to text

Yukti ID

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